WORKED SOLUTION

STEP II 2009, Question 6

Pure • Bounding Infinite Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q6

The key idea

We can't sum $\sum 1/F_i$ exactly, but we can trap it. The recurrence gives two easy comparisons: each Fibonacci number is less than twice the one before it, and more than twice the one two places before it. The first comparison bounds the tail of the series below by a geometric series. The second relates the tail to a shifted copy of itself, which bounds it above. Adding a few terms exactly before using either bound tightens the estimates as much as we like.

Solution

Adding successive pairs gives

$$F_3 = 2, \quad F_4 = 3, \quad F_5 = 5, \quad F_6 = 8, \quad F_7 = 13, \quad F_8 = 21, \quad F_9 = 34, \quad F_{10} = 55.$$

All the F_n are positive, and $F_{n+1} = F_n + F_{n-1} > F_n$ for $n \geq 2$, so the sequence is strictly increasing from F_2 onwards.

The series converges. For $i \geq 4$ we have $F_i = F_{i-1} + F_{i-2} > 2F_{i-2}$, since $F_{i-1} > F_{i-2}$ when $i - 2 \geq 2$. As $F_1 = 1 \geq 2^{-1/2}$ and $F_2 = 1 \geq 2^0$, while $F_3 = 2 \geq 2^{1/2}$, induction in steps of two gives $F_i \geq 2^{(i-2)/2}$ for every i. So $0 < 1/F_i \leq (1/\sqrt{2})^{i-2}$, and S converges by comparison with a geometric series of ratio $1/\sqrt{2} < 1$. This lets us manipulate the sums below freely. We'll also use one fact repeatedly. If $a_i \leq b_i$ for every i, and $a_i < b_i$ for at least one i, then $\sum a_i < \sum b_i$ for convergent series.

Part (i)

The lower bound. For $i \geq 4$, $F_{i-2} < F_{i-1}$, so

$$F_i = F_{i-1} + F_{i-2} < 2F_{i-1} \qquad \Longrightarrow \qquad \frac{1}{F_i} > \frac{1}{2F_{i-1}} \qquad (i \geq 4),$$

which is the first result. (At $i = 3$ we get equality, $F_3 = 2F_2$, which is why the inequality starts at $i = 4$.)

Applying this repeatedly, for $i \geq 4$,

$$\frac{1}{F_i} > \frac{1}{2F_{i-1}} > \frac{1}{2^2F_{i-2}} > \cdots > \frac{1}{2^{i-3}F_3} = \frac{1}{2^{i-2}}.$$

Summing over $i \geq 4$,

$$\sum_{i=4}^{\infty} \frac{1}{F_i} > \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = \frac{1}{2},$$

so

$$S = 1 + 1 + \tfrac{1}{2} + \sum_{i=4}^{\infty} \frac{1}{F_i} > \tfrac{5}{2} + \tfrac{1}{2} = 3.$$

The upper bound. Now use the other comparison: $1/F_i < 1/(2F_{i-2})$ for $i \geq 4$, from $F_i > 2F_{i-2}$ above. We apply it to every term from $i = 5$ onwards and set $j = i - 2$:

$$\sum_{i=5}^{\infty} \frac{1}{F_i} < \frac{1}{2} \sum_{i=5}^{\infty} \frac{1}{F_{i-2}} = \frac{1}{2} \sum_{j=3}^{\infty} \frac{1}{F_j} = \frac{1}{2}(S - 2).$$

The left-hand side is S minus its first four terms, $1 + 1 + \frac{1}{2} + \frac{1}{3} = \frac{17}{6}$. Hence

$$S - \tfrac{17}{6} < \tfrac{1}{2}S - 1 \qquad \Longrightarrow \qquad \tfrac{1}{2}S < \tfrac{11}{6} \qquad \Longrightarrow \qquad S < \tfrac{11}{3} = 3\tfrac{2}{3}.$$

Ask yourself Why start the upper-bound comparison at $i = 5$ rather than at $i = 4$?

Starting at $i = 4$ works but gives $S - \frac{5}{2} < \frac{1}{2}(S - 1)$, that is $S < 4$, which isn't good enough. The more terms we add exactly before estimating, the less the factor $\frac{1}{2}$ costs us. That idea drives part (ii).

Part (ii)

Write $S_k = \sum_{i=1}^{k} 1/F_i$ for the partial sums and $T_k = \sum_{i=k}^{\infty} 1/F_i$ for the tails, so $S = S_{k-1} + T_k$. Part (i) used two general bounds, both valid for $k \geq 3$.

Lower: exactly as before, $1/F_i > 1/(2^{i-k}F_k)$ for $i > k$, so

$$T_{k+1} > \frac{1}{F_k}\left(\tfrac{1}{2} + \tfrac{1}{4} + \cdots\right) = \frac{1}{F_k}, \qquad \text{so} \qquad S > S_k + \frac{1}{F_k}.$$

Upper: $1/F_i < 1/(2F_{i-2})$ for $i \geq k + 2 \geq 5$ gives $T_{k+2} < \frac{1}{2}T_k$, and $T_{k+2} = T_k - \frac{1}{F_k} - \frac{1}{F_{k+1}}$, so

$$T_k < 2\left(\frac{1}{F_k} + \frac{1}{F_{k+1}}\right), \qquad \text{so} \qquad S < S_{k-1} + 2\left(\frac{1}{F_k} + \frac{1}{F_{k+1}}\right).$$

With $k = 3$ these are the bounds 3 and $3\frac{2}{3}$ of part (i). Now take $k = 5$ in both.

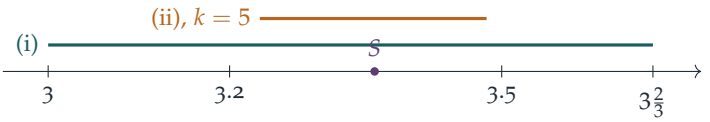
For the lower bound, $S_5 = 1+1+\frac{1}{2}+\frac{1}{3}+\frac{1}{5} = \frac{91}{30}$, so

$$S > \frac{91}{30} + \frac{1}{5} = \frac{97}{30} > \frac{96}{30} = 3.2.$$

For the upper bound, $S_4 = \frac{17}{6}$ and $F_5 = 5$, $F_6 = 8$, so

$$S < \frac{17}{6} + 2\left(\frac{1}{5}+\frac{1}{8}\right) = \frac{17}{6} + \frac{13}{20} = \frac{170+39}{60} = \frac{209}{60} < \frac{210}{60} = 3.5.$$

Together these give $3.2 < S < 3.5$, as required. ($k = 4$ isn't enough: it gives only $\frac{19}{6} < S < \frac{107}{30}$, about $3.17 < S < 3.57$.)



Checks

- Adding up the first 200 terms numerically gives $S \approx 3.35989$. This lies inside both intervals, and the bounds for $k = 3, \ldots, 6$ all bracket it: $3 < \frac{19}{6} < \frac{97}{30} < \frac{197}{60} < S < \frac{2681}{780} < \frac{209}{60} < \frac{107}{30} < \frac{11}{3}$.
- Test the comparisons on actual values: $F_7 = 13 < 2 \times 8$ and $13 > 2 \times 5$.

Common slips

- Using $1/F_i > 1/(2F_{i-1})$ at $i = 3$, where it's an equality. That breaks the strictness, or produces the wrong geometric sum.
- Manipulating S as a number ("$S - \frac{17}{6} < \frac{1}{2}S - 1$") without first knowing that the series converges.
- Giving up on the upper bound because the comparison started at $i = 4$ only gives $S < 4$. Add more terms exactly before estimating.

Beyond STEP

The value is $S \approx 3.3598856\ldots$, the *reciprocal Fibonacci constant*. It is known to be irrational (a result of André-Jeannin, 1989), but no closed form is known. The comparisons here are ratio tests in disguise. F_{i+1}/F_i tends to the golden ratio $\varphi = \frac{1}{2}(1+\sqrt{5})$, so the tail behaves like a geometric series with ratio $1/\varphi \approx 0.618$, between our crude ratios $\frac{1}{2}$ and $1/\sqrt{2}$. Comparison tests and tail estimates like these are core to a first analysis course. Abbott's *Understanding Analysis* is an excellent place to meet them properly.

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